ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** particles	Tiny particles.
2	**(c)** freezing	Fusion reverse.
3	**(c)** fit into the spaces	Gaps.
4	**(c)** have spaces between them	Spaces present.
5	**(c)** sublimation	Solid to gas.
6	**(b)** diffusion	Mixing.
7	**(c)** fixed shape	Definite shape.
8	**(a)** solids	Vibrate in position.
9	**(b)** faster than cold water	Fast particles.
10	**(b)** fixed volume	Definite volume.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tiny particles (2).	2
12	Liquid changing to vapour below boiling point (2).	2
13	0 degrees Celsius (2).	2
14	Mixing of particles on their own (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Fixed shape (1); Fixed volume (1); Cannot be compressed (1).	1+1+1
16	Drying clothes (1); Drying puddles (1); Sweat drying (1).	1+1+1
17	Sweat (1); Earthen pots (1); Coolers (1).	1+1+1
	OR: No fixed shape (1); Fixed volume (1); Flow (1).	1+1+1
18	Dew (1); Fog (1); Droplets on a cold glass (1).	1+1+1
	OR: Particles hit walls (1); Collisions give force (1); More collisions more pressure (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Tiny particles (1); Spaces between them (1); Constant motion (1); Forces of attraction (1); Examples (1).	1+1+1+1+1
	OR: Ice warms to 0 (1); Melts at 0 (1); Water warms to 100 (1); Boils at 100 (1); Steam warms (1).	1+1+1+1+1
20	Solid: close and fixed (1); Liquid: close, move (1); Gas: far, free (1); Forces strong, medium, weak (1); Examples (1).	1+1+1+1+1
	OR: Fill container (1); Compressible (1); Pressure by collisions (1); More at high temperature (1); Examples (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** gases	Gas.
2	**(b)** gaseous state	Vapour.
3	**(d)** no fixed shape	Fill the container.
4	**(c)** gas	Gaps large.
5	**(a)** solid state	Frozen.
6	**(a)** diffusion	Diffusion.
7	**(b)** sun and wind	Evaporation.
8	**(d)** evaporation	More moisture, slower.
9	**(d)** 0 degrees Celsius	Melting point.
10	**(b)** cooling	Cooling.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Vapour changing to liquid (2).	2
12	Particles move freely (1); Spread out in all directions (1).	1+1
13	Solid, liquid (1); Gas (1).	1+1
14	Solid to gas directly (1); Camphor (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	No fixed shape (1); No fixed volume (1); Easily compressed (1).	1+1+1
16	Melting (1); Boiling or evaporation (1); Condensation (1).	1+1+1
17	Particles move faster (1); Spaces increase (1); May change state (1).	1+1+1
	OR: Smell of perfume (1); Ink in water (1); Smell of food (1).	1+1+1
18	Temperature (1); Surface vs whole liquid (1); Slow vs fast (1).	1+1+1
	OR: Boiling water (1); Drying clothes (1); Cooking steam (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Mixing of particles (1); Example: ink (1); Faster in gases (1); Faster when hot (1); Evidence of motion (1).	1+1+1+1+1
	OR: Gas (1); Compressed (1); Cooled (1); Liquid form (1); Easy to carry (1).	1+1+1+1+1
20	Melting (1); Freezing (1); Boiling (1); Condensation (1); Sublimation (1).	1+1+1+1+1
	OR: Fast particles leave (1); Remaining cool down (1); Sweat (1); Earthen pot (1); Acetone on hand (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **Particulate Nature of Matter** | Class 8 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** degree Celsius	degC.
2	**(c)** 100 degrees Celsius	Boiling point.
3	**(a)** condensation	Water vapour.
4	**(c)** liquefied gas	Under pressure.
5	**(c)** vaporisation	Becomes vapour.
6	**(b)** carbon dioxide	Sublimes.
7	**(a)** vapour	Gas.
8	**(a)** condensation	On cooling.
9	**(c)** liquid	Room temperature.
10	**(b)** temperature	Temperature.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Water evaporates (1); Heat speeds it (1).	1+1
12	Solid changes to liquid on heating (2).	2
13	100 degrees Celsius (2).	2
14	Air (1); LPG gas (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Temperature (1); Surface area (1); Wind (1).	1+1+1
16	Water evaporates (1); Takes heat (1); Floor cools (1).	1+1+1
17	Oxygen: gas (1); Iron: solid (1); Milk: liquid (1).	1+1+1
	OR: Camphor (1); Naphthalene (1); Dry ice (1).	1+1+1
18	Pores let water seep out (1); It evaporates (1); Causes cooling (1).	1+1+1
	OR: Particles are spaced widely in ice (1); Ice is less dense (1); So it floats (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	At any temperature (1); At surface (1); Causes cooling (1); Factors (1); Examples (1).	1+1+1+1+1
	OR: Solid to gas (1); Camphor (1); Naphthalene (1); Dry ice (1); Used to separate mixtures (1).	1+1+1+1+1
20	Temperature high (1); Particles move faster (1); More evaporation (1); Wind helps (1); Humidity low (1).	1+1+1+1+1
	OR: Shape (1); Volume (1); Compressibility (1); Arrangement (1); Forces (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that particles of a solid are at rest; they vibrate about fixed positions.
- Mixing up the spacing: gas particles are far apart, liquid and solid particles are close.
- Saying temperature keeps rising during melting or boiling; it stays constant during a change of state.
- Writing that evaporation happens only at the boiling point, or that it heats the surroundings; it causes cooling.
- Confusing diffusion (mixing of particles by themselves) with dissolving or with stirring.
- Stating that a liquid has a definite shape; it has a definite volume only.

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